



Economics

Macroeconomics Equilibrium

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GCE ADVANCED LEVEL ENGLISH MEDIUM

making of an economy as a whole. For example, using interest rates, taxes and government spending to regulate an economy's growth and stability

Macroeconomic variables

- Aggregate output
- Employment
- General price level
- Exchange rate
- Interest rate
- Balance of payments

Macroeconomic objectives

- Full employment
- Economic stability
- Equity
- Economic growth
- Sustainable development

Macroeconomic problems

- Economic recessions and economic expansions
- Unemployment
- Inflation

Macroeconomic policies

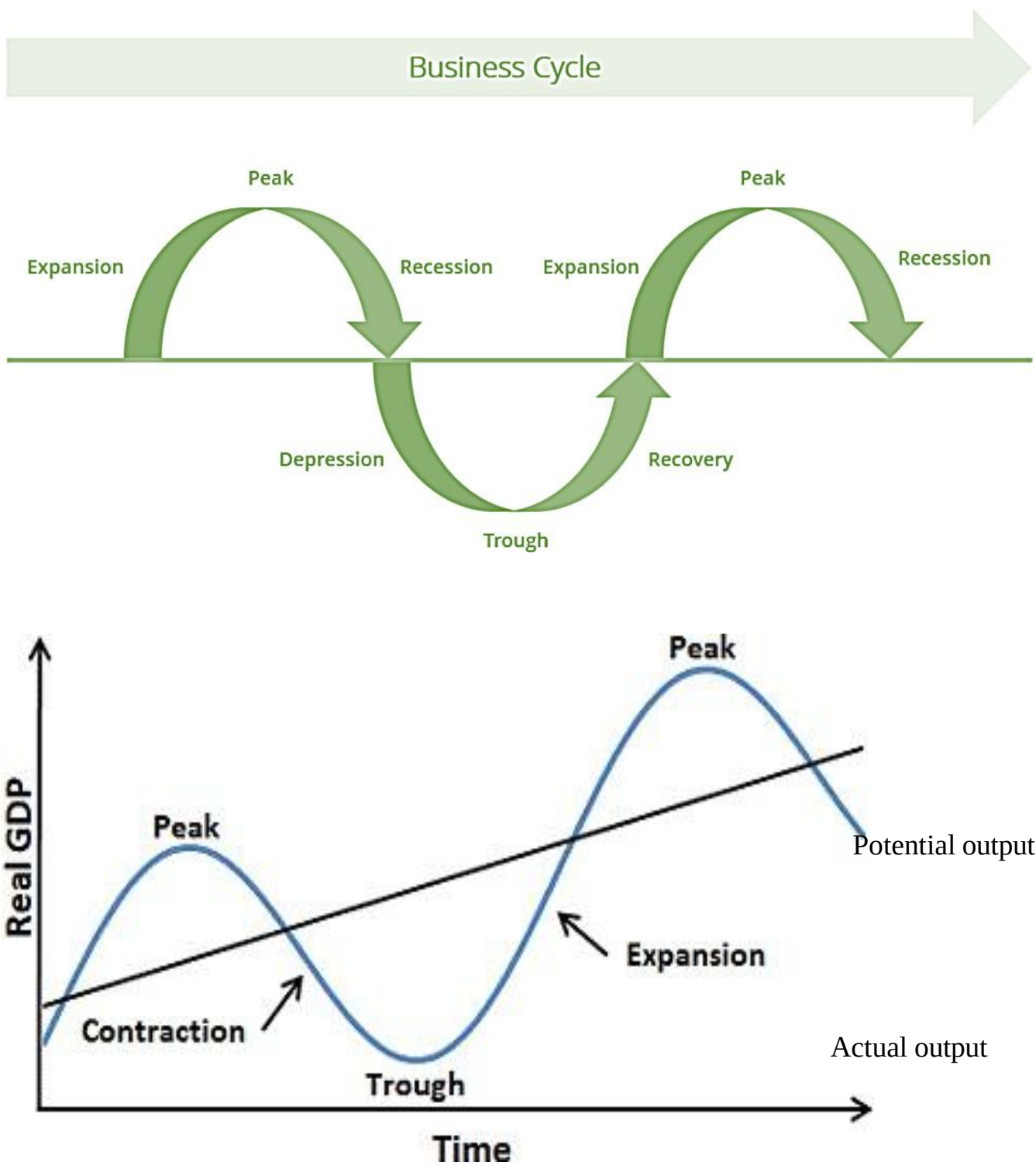
- Monetary policies
- Fiscal policies
- Supply side policies
- Income policies
- Direct control policies
- Foreign trade policies

Special Notes

Business cycle

The business cycle can be defined as the downward and upward fluctuations of gross domestic product (GDP) along with its natural growth rate over a long period of time.

⊛ This business cycle also known as Economic cycle / Trade cycle.



✧ Potential output & Actual output

In economics, potential output refers to the highest level of real gross domestic product that can be sustained over the long term. The long run behavior of aggregate economy is shown by the potential output.

Actual output happens in real life while potential output shows the level that could be achieved. Short run behavior of aggregate economy is shown by the actual output.

✧ Trough

The lowest situation of actual output below potential output is called trough.

✧ Peak

the highest situation of actual output above the potential output is called peak.

✧ Expansion

The stage which moves from trough to peak is called expansion.



✧ Recession

The stage which moves from peak to trough is called recession.



Macroeconomic Equilibrium

Different sectors contributing to macroeconomic activities

1. Household sector

Household sector owns factors of production and providing these factors of production to the business sector and earn factor income.

2. Business sector

Purchasing of goods and services produced by the business sector.

3. Government sector

Household sector pays taxes to the government and consume public services provided by the government.

4. Foreign sector

The following illustration can be used to show the interrelationship among the agents.

Macroeconomic Equilibrium

Macroeconomic equilibrium is a condition in the economy in which the quantity of aggregate demand equals the quantity of aggregate supply. If there are changes in either aggregate demand or aggregate supply, you could also see a change in price, unemployment, and inflation. For example, if the aggregate demand for your lemonade is too low, then your new business venture won't need to keep making as much lemonade and if you hired any friends to help you run your lemonade stand, you may have to let them go. This is because if customers are not buying lemonade, you won't be making money, which means you won't be able to pay any of your friends. When this happens at large companies, workers are often laid off, which ultimately causes the unemployment rate to increase.

Approaches that can be taken to represent macroeconomic equilibrium...

1 . Ad = As Method - _____

2. Y = E Method - _____

3. W=J Method - _____

1 . Ad = As Method - _____

Aggregate Supply is the total quantity of all goods and services produced in an economy at all possible price levels at a given time.

* _____

Aggregate Demand is the total quantity of all goods and services consumed in an economy at all possible price levels at a given time.

* _____

2. Y = E Method - _____

Aggregate Income is the sum of all income earned by resource suppliers in an economy during a given time period. In other words sum of the value added at each stage of production.

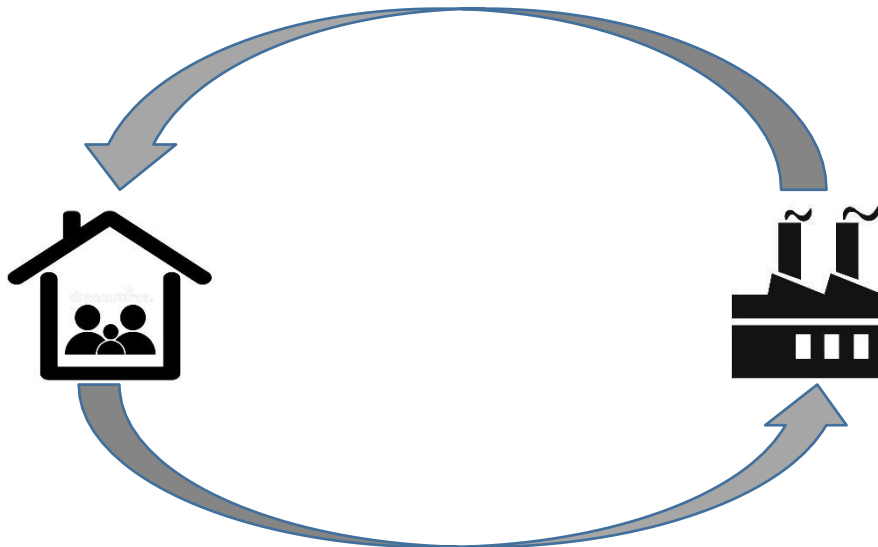
Aggregate Income = GDP = Aggregate Expenditure

The **aggregate income** is the total amount of income that is generated by all people, businesses, and government in a given country.

- E = Employee income
- B = Business owner income
- R = Rental income
- C = Corporate income
- I = Interest income
- G = Government income
- S = Government subsidies

To calculate the aggregate income, we use this formula: $E + B + R + C + I + (G - S) = \text{aggregate income}$.

***Remember that we begin by subtracting government subsidies from the government income, then add the difference to all other variables**



★ Aggregate expenditure

The aggregate expenditure is defined as the expenditure which is spent for goods and services by economic agents in a certain period of time.



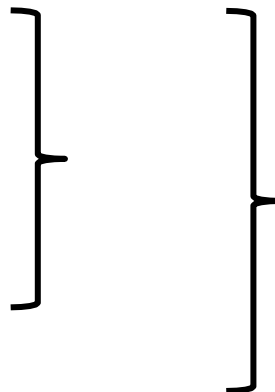
components of aggregate expenditure

☐ Consumption Expenditure (C)

☐ Investment Expenditure (I)

☐ Government Purchases (G)

☐ Net Exports (Nx) (X-M)



★ Consumption expenditure

Consumption expenditure is the expenditure which is spent for consumption of goods and services by households.

Function of consumption

☆ $C = b(Y_d)$



Without Autonomous consumption

C = Household consumption

☆ $C = a + bY_d$



With Autonomous consumption

a = Autonomous consumption
b = Marginal propensity to consume
Y_d = Disposable Income

The Autonomous consumption is demonstrated by the coefficient of "a" with the consumption function. The consumption expenditure in zero income is known as autonomous consumption. Namely, consumption which is determined without the effect of income.

The marginal Propensity to consume is demonstrated by the coefficient of "b" with the consumption function. The percentage change in consumption relative to the percentage change in disposable income is the marginal propensity to consume. It is the quantity changed in the consumptions when income changed by one unit.

MPC Equation

Y	C
150	280
280	360

☆ Calculate consumption function by using the table as follows

01--

1- Calculate MPC

 ΔY

=

MPC (b) = $\frac{\Delta C}{\Delta Y}$
-Savings-

Saving is the part of income which is not spend on consumption.

☆ Savings function

 $S = Y - C$


C = Household Savings

Y = Income

S = consumption expenditure

☆ Derive the savings function by using the consumption function
 $S = Y - C$
 $S = Y - (a + by)$

EX-

 $C = 100 + 0.8 Y$
 $S = Y - (100 + 0.8Y)$
 $S = Y - 100 - 0.8Y$
 $S = -100 + 0.2 Y$

EX-

$$C = 200 + 0.75Y$$

$$S = Y - (200 + 0.75Y)$$

$$S = Y - 200 + 0.75Y$$

$$S = -200 + 0.25 Y$$

* Find the savings function by using the consumption function.

1- $C = 150 + 0.75 Y_d$

2- $C = 300 + 0.65 Y_d$

3- $C = 200 + 0.85 Y_d$

$$C = 1350 + 0.7 Y_d$$

4- $C = 1000 + 0.75 Y_d$

5- $C = 1800 + 0.8 Y_d$

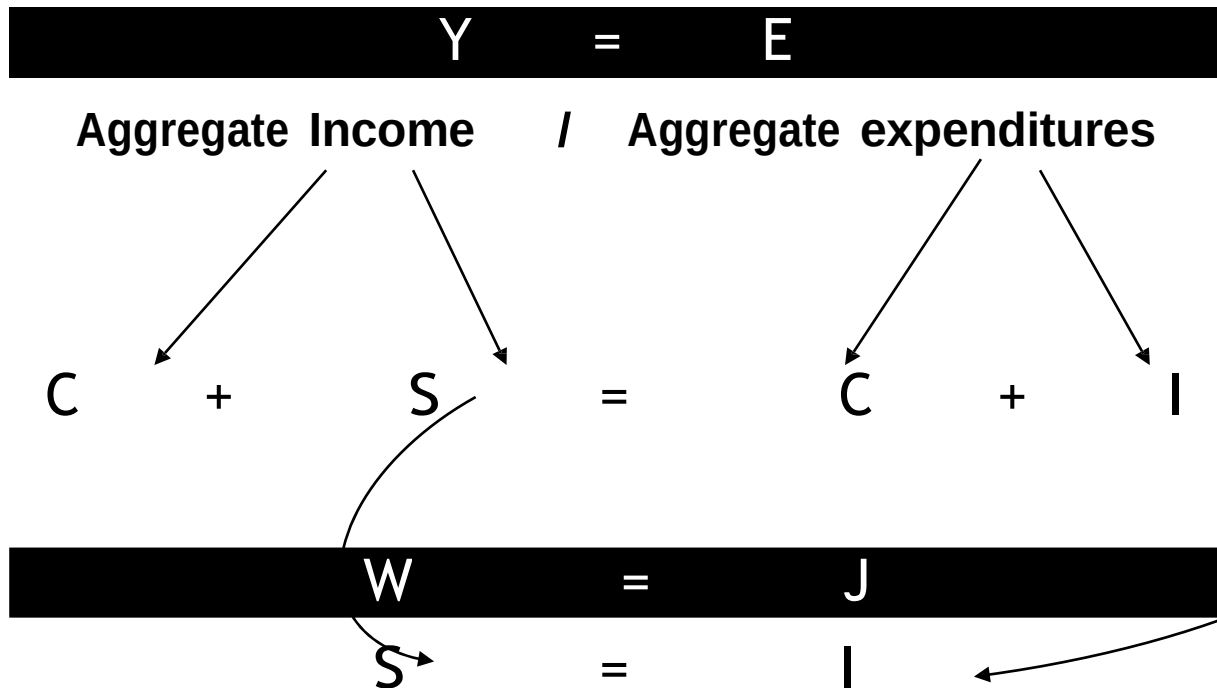
6- $C = 280 + 0.85 Y_d$

7- $C = 650 + 0.75 Y_d$

8- $C = 190 + 0.75 Y_d$

9- $C = 150 + 0.8 Y_d$

$$10- C= 300+ 0.75 Y_d$$



Macroeconomic Equilibrium of a closed economy

★ A closed economy is one that has no trading activity with outside economies.

The closed economy is therefore entirely self-sufficient, which means no imports come into the country and no exports leave the country.

The goal of a closed economy is to provide domestic consumers with everything they need from within the country's borders.

Household sector

Business sector

Government

$$Y = E$$

Aggregate Income / Aggregate expenditures

$$C + S + T = C + I + G$$

$$J = W$$

$$I + G = S + T - R$$

Y = E method
Y = C + I + G

$$c = 100 + 0.75 Y_d$$

$$I = 50$$

$$G = 200$$

$$S = -100 + 0.25 Y_d$$

$$Y = C + I + G$$

$$= 100 + 0.75Y + 50 + 200$$

$$= 350 + 0.75Y$$

$$Y - 0.75Y = 350$$

$$0.25 = \frac{350}{0.25}$$

$$0.25 \quad 0.25$$

$$Y = 1400$$

W = J method

$$S + T - R = I + G$$

$$c = 100 + 0.75 Y_d$$

$$I = 50$$

$$G = 200$$

$$S = -100 + 0.25 Y_d$$

$$-100 + 0.25Y = 50 + 200$$

$$0.25Y = 50 + 200 + 100$$

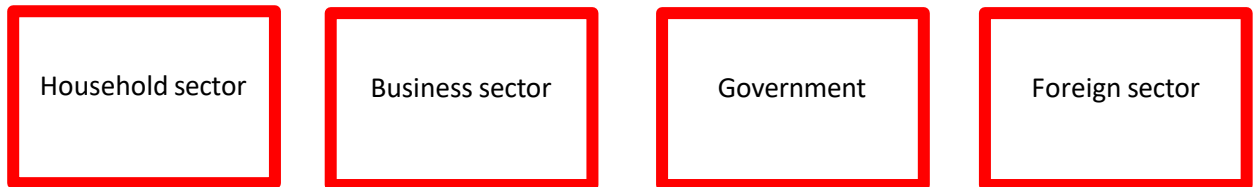
$$0.25 Y = 350$$

$$0.25 \quad 0.25$$

$$Y = 1400$$

Macroeconomic Equilibrium of an Open economy

✪ An open economy where goods and services are traded with other countries without rules or limits.



$$Y = E$$

Aggregate Income / Aggregate expenditures

$$Y = C + I + G + X - M$$

$$J = W$$

$$I + G + X = S + T - R + M$$

- Consumption = $200 + 0.8Y_d$
- Investments = 100
- Government purchasers = 200
- Taxes = 50
- Net exports = -100

$$\begin{aligned}
 Y_d &= Y - t \\
 Y &= E \\
 Y &= C + I + G + (X - M) \\
 Y &= 200 + 0.8(Y - t) + 200 + 100 + (-100) \\
 Y &= 400 + 0.8(Y - 50) \\
 Y &= 400 + 0.8Y - 40 \\
 Y - 0.8Y &= 360 \\
 0.2Y &= 360 \\
 Y &= 360 \div 0.2 \\
 Y &= 1800
 \end{aligned}$$

If the investment in the above example is increased by 100, the new equilibrium can be calculated as below.

$$Y = E$$

$$Y = C + I + G + (X - M)$$

$$Y = 200 + 0.8(Y - T) + 200 + 200 + (-100)$$

$$Y = 500 + 0.8(Y - 50)$$

$$Y = 500 + 0.8Y - 40$$

$$Y = 460$$

$$0.2$$

$$Y = 2300$$

★ Macroeconomic Multiplier

Change of equilibrium due to the increase or decrease in autonomous expenditures is determined by the largeness of the multiplier.

The coefficient that shows the amount of increase in equilibrium output with an increase of the autonomous expenditures of an economy by a particular amount is known as multiplier.

★ The multiplier of a simple economy can be presented as below.

$$K = \frac{1}{1 - b} = \frac{1}{1 - MPC} = \frac{1}{MPS}$$

$$01^* C = 200 + 0.8Y$$

$$K = \frac{1}{1 - MPC}$$

$$K = \frac{1}{1 - 0.8}$$

$$K = \frac{1}{0.2}$$

$$= 5$$

$$02^* \quad C = 400 + 0.9Y$$

$$K = \frac{1}{1 - MPC}$$

$$K = \frac{1}{1 - 0.9}$$

$$K = \frac{1}{0.1}$$

$$= 10$$

$$03^* \quad C = 800 + 0.75Y$$

$$K = \frac{1}{1 - MPC}$$

$$K = \frac{1}{1 - 0.75}$$

$$K = \frac{1}{0.25}$$

$$= 4$$

$$04^* \quad C = 1700 + 0.99Y$$

$$K = \frac{1}{1 - MPC}$$

$$K = \frac{1}{1 - 0.99}$$

$$K = \frac{1}{0.01}$$

$$=$$

01- When government expenditures are increased by Rs. 50 million under the $C = 100 + 0.8Y_d$ consumption function. Calculate the economy expands of the Equilibrium income.

$$K = \frac{\Delta Y}{\Delta E}$$

$$\Delta Y = K \ll \Delta E$$

$$Y_2 = Y_1 + \Delta Y$$

$$\begin{aligned}\Delta Y &= K * \Delta E \\ &= 5 * 50 \\ &= 250\end{aligned}$$

02- Example When government expenditures are increased by Rs. 100 million under the $C=500+0.75Y_d$ consumption function. Calculate the economy expands of the Equilibrium income.

$$K = \frac{\Delta Y}{\Delta E}$$

$$\Delta \mathbf{Y} = \mathbf{K} \ll \Delta \mathbf{E}$$

$$\mathbf{Y}_2 = \mathbf{Y}_1 + \Delta \mathbf{Y}$$

$$\Delta Y = K * \Delta E$$

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